Title: Login Behavior Analysis Security System

Student Name: Talluri Hima Bindu Sree
Student ID: 2520030484
Cluster: 1
Section: 1

Under the Guidance
of

Supervisor Name: Jayaram Boga
Assistant Professor
Department of CSE

(DEEMED TO BE UNIVERSITY)

OUTLINE

1. Abstract
2. Existing System
3. Proposed System
4. System Architecture
5. Project Modules
6. Objectives
7. Co's Attainments - FWD
    1. Co1
    2. Co2 …….. 5. Co5
8. Co's Attainments - DSA
    1. Co1
    2. Co2 …….. 5. Co5
9. Screen Shots for FWD & DSA
10. Conclusion
11. Geo-Tagged Photos

Abstract

- The Login Behavior Analysis Security System is a web-based security application designed to analyze user login behavior and detect suspicious activities using rule-based analysis.
- Instead of relying only on credential verification, the system monitors behavioral patterns such as unusual login times, multiple login attempts within a short period, and repeated failed login attempts. When abnormal behavior is detected, the system flags the activity and provides a clear explanation of the rules that were triggered, improving transparency in security decisions.
- This project serves as a lightweight and efficient model that demonstrates the internal working of real-world login security systems, focusing on clarity, transparency, and educational value.

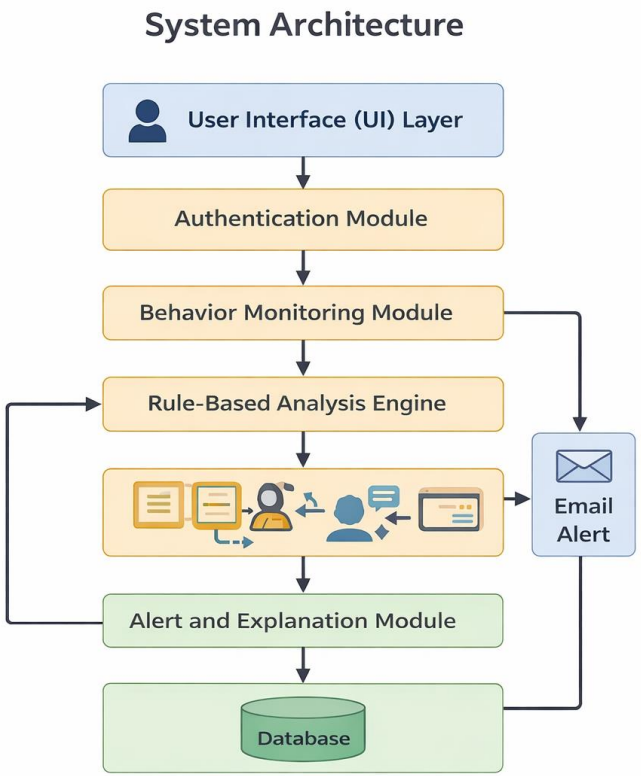
Existing System

- Enhances traditional login systems by analyzing user login behavior in addition to credential verification.
- Monitors patterns such as unusual login times, repeated failed logins, and multiple login attempts in a short period.
- Uses predefined rule-based analysis to evaluate each login attempt.
- Flags suspicious activities when abnormal behavior is detected.
- Records login attempts and security events for monitoring and analysis.
- Displays the specific rule that triggered the alert to ensure transparency.
- Provides a simple, efficient, and explainable behavior-based login security model for learning and demonstration purposes.

Proposed System

- Enhances traditional login by checking credentials and user behavior.
- Monitors unusual login times, multiple attempts in short periods, and repeated failed logins.
- Uses predefined rules to analyze each login attempt.
- Flags suspicious activity when rules are violated.
- Records login attempts for monitoring and analysis.
- Displays the rule that triggered the alert for transparency.
- Helps administrators understand why a login was flagged.
- Provides a simple, efficient, and explainable behavior-based security model for educational and demonstration purposes.

System Architecture



- ✓ **User Login Module** – Designed the login interface using HTML and CSS with proper webpage structure
- ✓ **Authentication Module** – Implemented HTML forms and responsive layouts for the security dashboard
- ✓ **COS Attainment** – Applied JavaScript logic, conditions, functions to process login data
- ✓ **Database** – Used DOM mapiulation and event handling

Project Modules

- **User Login Module**
  Allows users to enter their username and password through the login interface.
- **Authentication Module**
  Verifies the entered credentials and determines whether the login is successful or failed.
- **Behavior Monitoring Module**
  Records login details such as login time, number of attempts, and failed logins to track user behavior.
- **Rule-Based Analysis Module**
  Applies predefined rules to detect suspicious activities like unusual login times or repeated failed attempts.
- **Alert and Notification Module**
  Flags suspicious logins and displays alerts or sends email notifications when abnormal behavior is detected.
- **Data Storage Module**
  Stores login history, triggered rules, and security events for analysis and monitoring.

Objectives

- To develop a secure login system that goes beyond basic username and password authentication by monitoring user login behavior.
- To analyze user login patterns such as unusual login times, multiple login attempts within a short period, and repeated failed login attempts.
- To detect and flag suspicious login activities using predefined rule-based analysis.
- To provide transparency in security decisions by clearly explaining which rule triggered the suspicious activity alert.
- To maintain a record of login attempts for monitoring and analysis of user behavior.
- To create a simple and educational model that demonstrates how behavior-based security systems work in real-world applications.

Co’s Attainments - FWD

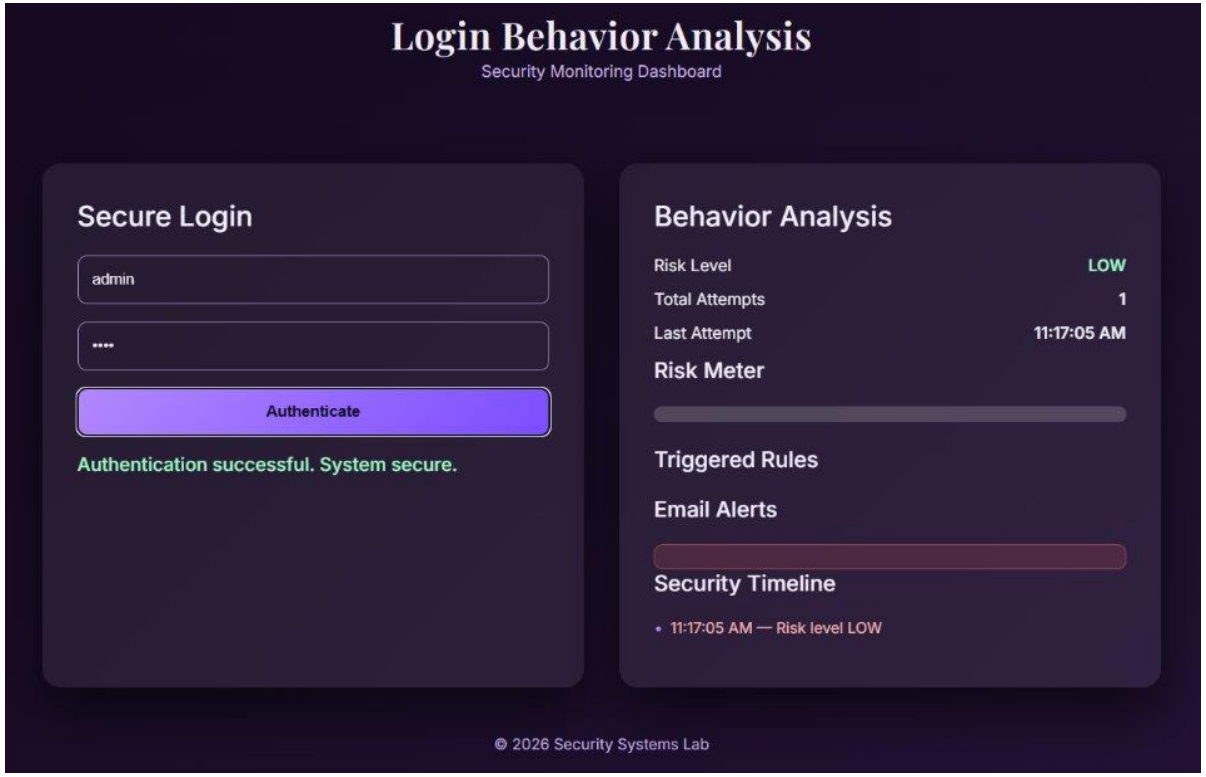
- **CO1 Attainment**
- Designed the login interface using HTML and CSS with proper webpage structure.
- **CO2 Attainment**
- Implemented HTML forms and responsive layouts for the security dashboard.
- **CO3 Attainment**
- Applied JavaScript logic, conditions, and functions to process login data.
- **CO4 Attainment**
- Used DOM manipulation and event handling to update risk levels and alerts dynamically.
- **CO5 Attainment**
- Developed a web-based security dashboard for analyzing and monitoring login behaviour.

Co’s Attainments - DSA

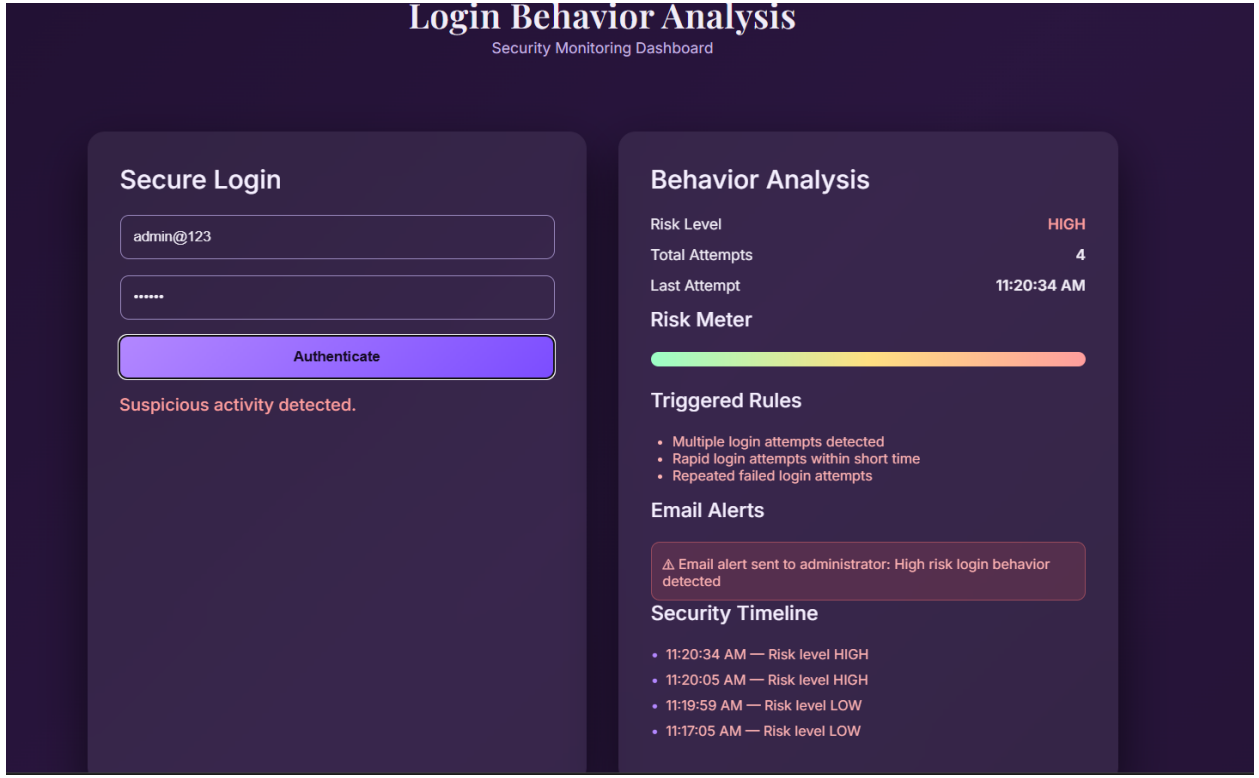
- **CO1 Attainment**
- Implemented Linear Search to identify failed login attempts.
- Used Bubble Sort to sort login attempts based on risk level.
- Analyzed time complexity of algorithms using Big-O notation.
- **CO2 Attainment**
- Implemented List ADT using ArrayList to store login history.
- Used LinkedList-based Queue to manage recent login attempts.
- Performed operations like insertion, traversal, and deletion on login data.
- **CO3 Attainment**
- Used Stack to track the most recent login attempt.
- Implemented Queue to maintain login attempts in FIFO order.
- Used PriorityQueue (Heap) to process high-risk login attempts first.

- **CO4 Attainment**
- Implemented HashMap (hash table) to track frequency of rule violations.
- Utilized Java Collections (List, Queue, Map, PriorityQueue) for efficient data handling.
- Enabled fast lookup and updates using hashing techniques.

Screenshots for FWD



Screenshots for FWD



Screenshots for DSA

```
Login Behavior Analysis System
Type 'exit' to stop

Enter Username: admin
Enter Password: 1234
Enter Time (HH:MM): 10:22
Enter Hour (0-23): 3
Login Risk Level: LOW
Enter Username: admin@123
Enter Password: 3434
Enter Time (HH:MM): 12
Enter Hour (0-23): 22
Login Risk Level: MEDIUM
Enter Username: user@34
Enter Password: 333
Enter Time (HH:MM): 31
Enter Hour (0-23): 1
Login Risk Level: HIGH
Enter Username: exit
```

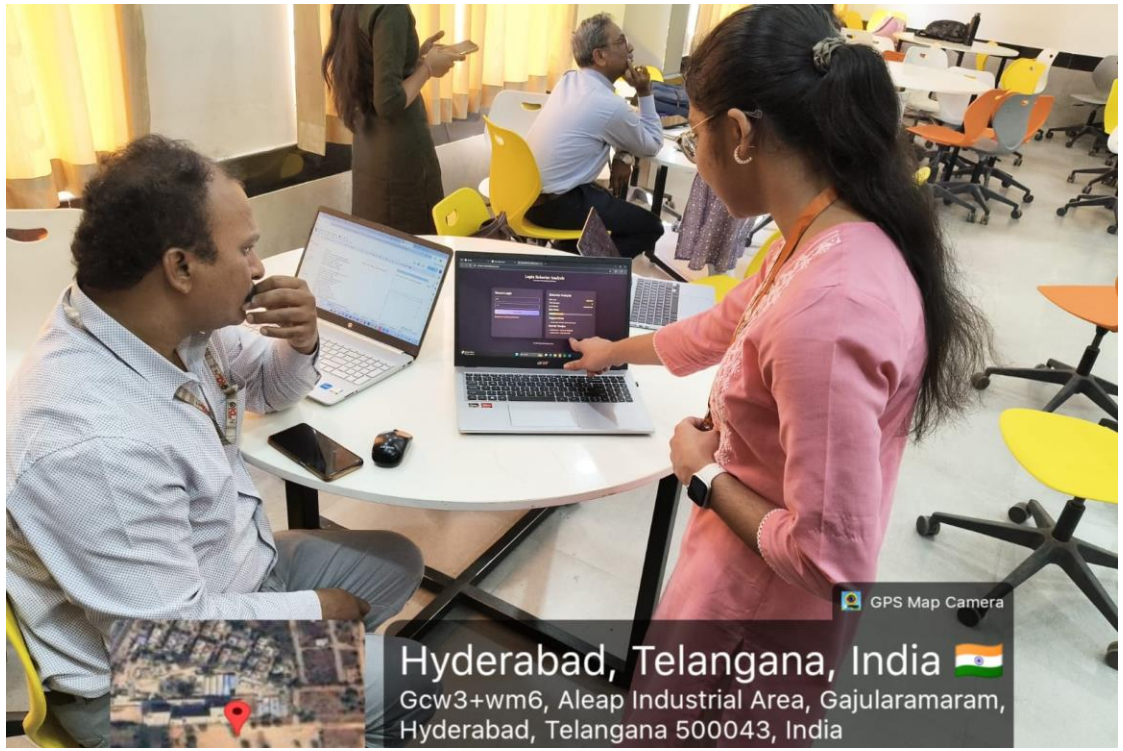
```
Most Recent Login (Stack):
Time        Success     Risk
--------------------------------
31          false       6

Failed Login Attempts:
Time        Success     Risk
--------------------------------
12          false       3
31          false       6

Login Attempts Sorted by Risk:
Time        Success     Risk
--------------------------------
10:22       true        1
12          false       3
31          false       6

Rule Frequency:
Rule                        Count
--------------------------------------
Repeated Failure            1
Failed Login                2
Unusual Login Time          2

High Risk Logins First:
Time        Success     Risk
--------------------------------
31          false       6
12          false       3
10:22       true        1

Total Login Attempts: 3
```

Conclusion

The Login Behavior Analysis Security System provides a simple and effective approach to enhancing login security by analyzing user behavior in addition to basic credential verification. The system monitors login patterns such as unusual login times, multiple login attempts within a short period, and repeated failed login attempts. Using rule-based analysis, it detects suspicious activities and flags them for further attention. A key feature of this system is its transparency, as it clearly identifies the rule that triggered the alert and explains why a login attempt is considered suspicious. This makes the system easy to understand and useful for demonstrating how behavior-based security mechanisms work. Overall, the project shows how login monitoring and rule-based analysis can improve security while maintaining clarity, efficiency, and educational value.

Geo- Tag Photos

Login Behavior Analysis Security System
GPS Map Camera
Hyderabad,Telangana,India
Dollar Dreams, Bowrampet, Hyderabad, Telangana 500043, India
Lat 17.547258, Long 78.404754
Saturday, 07/02/2026 09:51 AM GMT+05:30
Note : Captured by GPS Map Camera


GPS Map Camera
Hyderabad, Telangana, India
Gcw3+wm6, Aleap Industrial Area, Gajularamaram, Hyderabad, Telangana 500043, India

THANK YOU